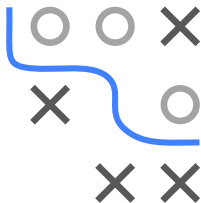


Deep dive

Gradient descent and optimality

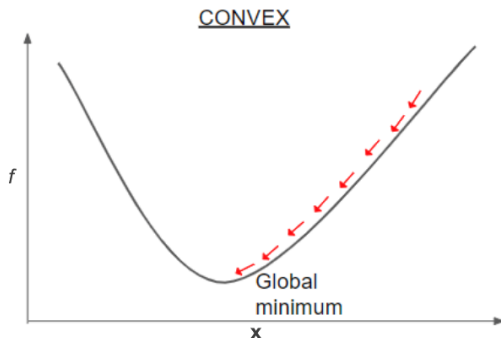
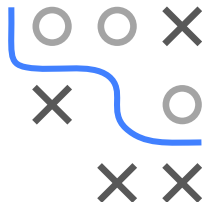


- Convergence of GD
- Proof strategy and tools
- Descent lemma



SETTING

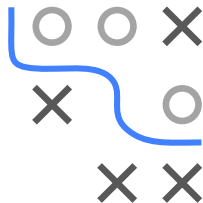
- GD is **greedy**: **locally optimal** moves in each iteration
- If f is **convex**, **differentiable** and has a **Lipschitz gradient**, GD converges to global minimum for sufficiently small step sizes.



SETTING

- **Assumptions:**

- f convex and differentiable
- Global minimum $\mathbf{x}^*$ exists
- f has Lipschitz gradient (∇f does not change too fast)



$$\|\nabla f(\mathbf{x}) - \nabla f(\tilde{\mathbf{x}})\| \leq L\|\mathbf{x} - \tilde{\mathbf{x}}\| \quad \text{for all } \mathbf{x}, \tilde{\mathbf{x}}$$

Theorem (Convergence of GD). GD with step size $\alpha \leq 1/L$ yields

$$f(\mathbf{x}^{[k]}) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}^{[0]} - \mathbf{x}^*\|^2}{2\alpha k}$$

In other words: GD converges with rate $\mathcal{O}(1/k)$.

PROOF STRATEGY

- ❶ Show that $f(\mathbf{x}^{[t]})$ **strictly decreases** with each iteration t

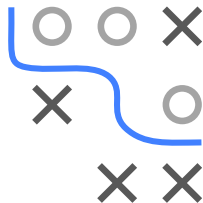
Descent lemma:

$$f(\mathbf{x}^{[t+1]}) \leq f(\mathbf{x}^{[t]}) - \frac{\alpha}{2} \|\nabla f(\mathbf{x}^{[t]})\|^2$$

- ❷ Bound **error of one step**

$$f(\mathbf{x}^{[t+1]}) - f(\mathbf{x}^*) \leq \frac{1}{2\alpha} \left(\|\mathbf{x}^{[t]} - \mathbf{x}^*\|^2 - \|\mathbf{x}^{[t+1]} - \mathbf{x}^*\|^2 \right)$$

- ❸ Finalize by **telescoping** argument

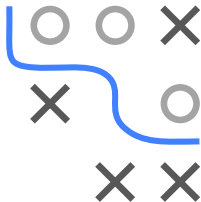
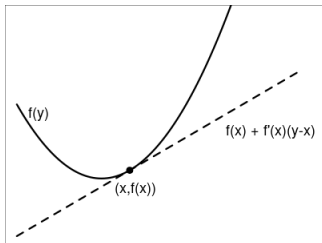


MAIN TOOL

- **Recall:** First order condition of convexity

Every tangent line of f is always below f .

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T (\mathbf{y} - \mathbf{x})$$



DESCENT LEMMA

- **Recall:** ∇f Lipschitz $\implies \nabla^2 f(\mathbf{x}) \preceq L \cdot I$ for all $\mathbf{x}$
- This gives convexity of $g(\mathbf{x}) := \frac{L}{2}\|\mathbf{x}\|^2 - f(\mathbf{x})$ since

$$\nabla^2 g(\mathbf{x}) = L \cdot I - \nabla^2 f(\mathbf{x}) \succeq 0$$

- First order condition of convexity of g yields

$$g(\mathbf{x}) \geq g(\mathbf{x}^{[t]}) + \nabla g(\mathbf{x}^{[t]})^T (\mathbf{x} - \mathbf{x}^{[t]})$$

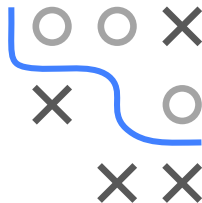
$$\Leftrightarrow \frac{L}{2}\|\mathbf{x}\|^2 - f(\mathbf{x}) \geq \frac{L}{2}\|\mathbf{x}^{[t]}\|^2 - f(\mathbf{x}^{[t]}) + (L\mathbf{x}^{[t]} - \nabla f(\mathbf{x}^{[t]}))^T (\mathbf{x} - \mathbf{x}^{[t]})$$

$$\Leftrightarrow \quad \quad \quad \vdots$$

$$\Leftrightarrow f(\mathbf{x}) \leq f(\mathbf{x}^{[t]}) + \nabla f(\mathbf{x}^{[t]})^T (\mathbf{x} - \mathbf{x}^{[t]}) + \frac{L}{2}\|\mathbf{x} - \mathbf{x}^{[t]}\|^2$$

- **Now:** One GD step with step size $\alpha \leq 1/L$:

$$\mathbf{x} \leftarrow \mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]})$$

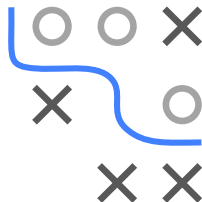


DESCENT LEMMA

$$\begin{aligned}f(\mathbf{x}^{[t+1]}) &\leq f(\mathbf{x}^{[t]}) + \nabla f(\mathbf{x}^{[t]})^T (\mathbf{x}^{[t+1]} - \mathbf{x}^{[t]}) + \frac{L}{2} \|\mathbf{x}^{[t+1]} - \mathbf{x}^{[t]}\|^2 \\&= f(\mathbf{x}^{[t]}) + \nabla f(\mathbf{x}^{[t]})^T (\mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) - \mathbf{x}^{[t]}) \\&\quad + \frac{L}{2} \|\mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) - \mathbf{x}^{[t]}\|^2 \\&= f(\mathbf{x}^{[t]}) - \nabla f(\mathbf{x}^{[t]})^T \alpha \nabla f(\mathbf{x}^{[t]}) + \frac{L}{2} \|\alpha \nabla f(\mathbf{x}^{[t]})\|^2 \\&= f(\mathbf{x}^{[t]}) - \alpha \|\nabla f(\mathbf{x}^{[t]})\|^2 + \frac{L\alpha^2}{2} \|\nabla f(\mathbf{x}^{[t]})\|^2 \\&\leq f(\mathbf{x}^{[t]}) - \frac{\alpha}{2} \|\nabla f(\mathbf{x}^{[t]})\|^2\end{aligned}$$

● **Note:** $\alpha \leq 1/L$ yields $L\alpha^2 \leq \alpha$

- $\|\nabla f(\mathbf{x}^{[t]})\|^2 > 0$ unless $\nabla f(\mathbf{x}) = \mathbf{0}$
- f **strictly decreases** with each GD iteration until optimum reached
- Descent lemma yields bound on **guaranteed progress** if $\alpha \leq 1/L$
(explains why GD may diverge if step sizes too large)



ONE STEP ERROR BOUND

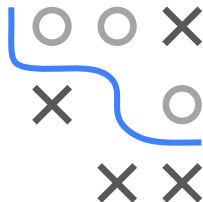
- Again, first order condition of convexity gives

$$f(\mathbf{x}^{[t]}) - f(\mathbf{x}^*) \leq \nabla f(\mathbf{x}^{[t]})^T (\mathbf{x}^{[t]} - \mathbf{x}^*).$$

- This and the descent lemma yields

$$\begin{aligned} f(\mathbf{x}^{[t+1]}) - f(\mathbf{x}^*) &\leq f(\mathbf{x}^{[t]}) - \frac{\alpha}{2} \|\nabla f(\mathbf{x}^{[t]})\|^2 - f(\mathbf{x}^*) \\ &= f(\mathbf{x}^{[t]}) - f(\mathbf{x}^*) - \frac{\alpha}{2} \|\nabla f(\mathbf{x}^{[t]})\|^2 \\ &\leq \nabla f(\mathbf{x}^{[t]})^T (\mathbf{x}^{[t]} - \mathbf{x}^*) - \frac{\alpha}{2} \|\nabla f(\mathbf{x}^{[t]})\|^2 \\ &= \frac{1}{2\alpha} \left(\|\mathbf{x}^{[t]} - \mathbf{x}^*\|^2 - \|\mathbf{x}^{[t]} - \mathbf{x}^* - \alpha \nabla f(\mathbf{x}^{[t]})\|^2 \right) \\ &= \frac{1}{2\alpha} \left(\|\mathbf{x}^{[t]} - \mathbf{x}^*\|^2 - \|\mathbf{x}^{[t+1]} - \mathbf{x}^*\|^2 \right) \end{aligned}$$

- **Note:** Line 3 $\rightarrow$ 4 is hard to see (just expand line 4).



FINALIZATION

- Summing over iterations yields

$$\begin{aligned}k(f(\mathbf{x}^{[k]}) - f(\mathbf{x}^*)) &\leq \sum_{t=1}^k [f(\mathbf{x}^{[t]}) - f(\mathbf{x}^*)] \\&\leq \sum_{t=1}^k \frac{1}{2\alpha} \left[\|\mathbf{x}^{[t-1]} - \mathbf{x}^*\|^2 - \|\mathbf{x}^{[t]} - \mathbf{x}^*\|^2 \right] \\&= \frac{1}{2\alpha} \left(\|\mathbf{x}^{[0]} - \mathbf{x}^*\|^2 - \|\mathbf{x}^{[k]} - \mathbf{x}^*\|^2 \right) \\&\leq \frac{1}{2\alpha} \left(\|\mathbf{x}^{[0]} - \mathbf{x}^*\|^2 \right).\end{aligned}$$

- Arguments:** Descent lemma (line 1).
Telescoping sum (line 2 $\rightarrow$ 3).

$$f(\mathbf{x}^{[t+1]}) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}^{[0]} - \mathbf{x}^*\|^2}{2\alpha k}$$

